



Honeypot Simulation

Personal Project 01

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1. Overview

This project simulates a honeypot environment to observe and analyse unauthorised SSH access attempts. The setup allows safe study of brute-force behaviour, command patterns, and attacker interaction in a controlled environment.

1.1 Problem

The project addresses the need for a safe and controlled environment to observe and analyse unauthorised SSH access attempts. This allows brute-force behaviour, credential attempts, command patterns, and attacker interaction to be studied without performing testing in a real-world environment.

1.2 Solution

Using the Cowrie honeypot on Ubuntu and an attacking Kali machine, the setup provides a controlled environment for capturing and analysing unauthorised SSH access attempts.

1.3 Goals

The goal of this project is to:

- Understand common brute-force and unauthorised access techniques.
- Capture and analyse attacker behaviour and commands.
- Gain hands-on experience with defensive security tools like Cowrie.

1.4 Components

Components	Purpose
Ubuntu 22.04 VM	Hosts Cowrie to capture and log attacker activity.
Kali Linux VM	Acts as an attacker performing SSH login attempts against the honeypot.
Cowrie	Provides a controlled, emulated SSH/Telnet environment and detailed logging of login attempts, commands, and interactions.
Hydra	Used to simulate SSH brute-force attacks.
Oracle VirtualBox	Hosts the two virtual machines used to create the isolated environment.

1.5 Features

- Simulates an SSH/Telnet service using the Cowrie honeypot.
- Captures SSH brute-force attempts (usernames, passwords).
- Logs all commands executed by attackers during sessions.
- Generates structured log files for analysis of attacker behaviour.

2. Scope

2.1 In Scope

- Setting up a honeypot environment using Cowrie on Ubuntu.
- Simulating SSH brute-force attacks from a Kali Linux machine.
- Capturing and logging attacker login attempts and commands.
- Basic analysis of attacker behaviour based on collected logs.

2.2 Out of Scope

- Development of custom scripts or programs for automated attack simulation or log analysis.
- Advanced malware analysis or payload execution.
- Integration with SIEM or large-scale monitoring systems.
- Simulation in a real-world environment.
- Defence beyond observation, such as automated blocking or response.

3. Setup & Design

3.1 Environment

The environment was built using two virtual machines inside Oracle VirtualBox.

- Ubuntu VM (Honeypot): Hosts Cowrie to capture and log attacker activity.
- Kali Linux (Attacker): Acts as an attacker performing SSH login attempts against the honeypot.

3.2 Operating System and Software

- Cowrie requires Python 3.10 or higher, so Ubuntu 22.04 was used as it meets the required version by default.
- Kali Linux was cloned from an existing setup.
- Cowrie was obtained from the official public GitHub repository and installed in Ubuntu VM.
- Required dependencies were installed, including Git, Python, and related libraries.

3.3 Network Design

- Both virtual machines were configured to use a Host-Only Adapter to allow direct communication while maintaining an isolated environment.
- The Ubuntu VM was temporarily configured to use a NAT Network while downloading the required dependencies before being switched back to the Host-Only Adapter.

3.4 Cowrie Design

Cowrie was used for the honeypot as it provides:

- A controlled, emulated SSH/Telnet environment.
- Detailed logging of all login attempts, commands, and interactions.

4. Methodology

4.1 Virtual Machine Setup

1. Ubuntu 22.04 virtual machine was installed and set up for the Cowrie honeypot.
2. An existing Kali Linux setup was cloned to be used as the attacker.
3. Both virtual machines were configured using a Host-Only Adapter to allow direct communication while maintaining an isolated environment.
4. The Ubuntu VM was temporarily set to NAT Network to download dependencies.

4.2 Installation of Cowrie Honeypot

Required system dependencies were installed:

```
sudo apt install python3-pip python3-dev libssl-dev libffi-dev build-essential
```

The Cowrie repository was cloned from the official GitHub source:

```
git clone https://github.com/cowrie/cowrie.git
```

4.3 Environment and Cowrie Setup

A Python virtual environment was created and activated:

```
python3 -m venv TESTenv  
source TESTenv/bin/activate
```

The Python package manager (pip) was updated and the required dependencies were installed:

```
pip3 install --upgrade pip
pip3 install -r requirements.txt
```

After installation, the Ubuntu VM was switched back to the Host-Only Adapter to maintain the isolated environment.

The Cowrie configuration file was initialised: `cp etc/cowrie.cfg.dist etc/cowrie.cfg`

4.4 Cowrie Configuration

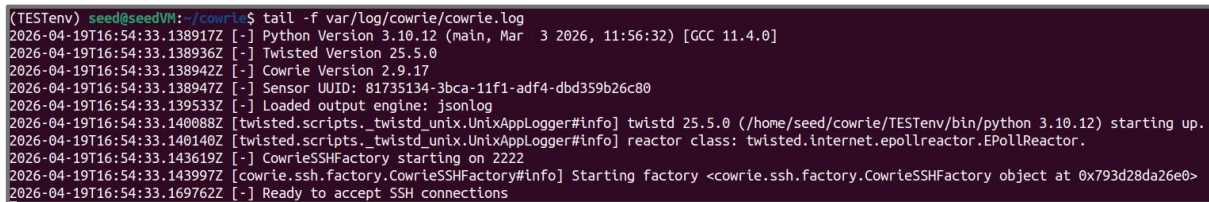
The default listening port was changed to allow connections from the Kali VM:

- Default: listen_endpoints = tcp:6415:interface=127.0.0.1
- Updated: listen_endpoints = tcp:2222:interface=0.0.0.0

4.5 Cowrie Execution and Monitoring

Cowrie was started using: `cowrie start`

Real-time log monitoring was performed using: `tail -f var/log/cowrie/cowrie.log`



```
(TESTenv) seed@seedVM:~/cowrie$ tail -f var/log/cowrie/cowrie.log
2026-04-19T16:54:33.138917Z [-] Python Version 3.10.12 (main, Mar  3 2026, 11:56:32) [GCC 11.4.0]
2026-04-19T16:54:33.138936Z [-] Twisted Version 25.5.0
2026-04-19T16:54:33.138942Z [-] Cowrie Version 2.9.17
2026-04-19T16:54:33.138947Z [-] Sensor UUID: 81735134-3bca-11f1-adf4-dbd359b26c80
2026-04-19T16:54:33.139533Z [-] Loaded output engine: jsonlog
2026-04-19T16:54:33.140088Z [twisted.scripts._twistd_unix.UnixAppLogger#info] twistd 25.5.0 (/home/seed/cowrie/TESTenv/bin/python 3.10.12) starting up.
2026-04-19T16:54:33.140140Z [twisted.scripts._twistd_unix.UnixAppLogger#info] reactor class: twisted.internet.epollreactor.EPollReactor.
2026-04-19T16:54:33.143619Z [-] CowrieSSHFactory starting on 2222
2026-04-19T16:54:33.143997Z [cowrie.ssh.factory.CowrieSSHFactory#info] Starting factory <cowrie.ssh.factory.CowrieSSHFactory object at 0x793d28da26e0>
2026-04-19T16:54:33.169762Z [-] Ready to accept SSH connections
```

Figure 1 – Cowrie Execution

4.6 Connectivity Verification

Note: IP addresses and port numbers have been masked in this document. The values shown are placeholders representing the actual testing environment used.

Network configuration:

- Ubuntu IP: 192.xxx.xx.xxx
- Kali IP: 192.xxx.xx.xxx

The SSH connection was tested from Kali using: `ssh root@192.xxx.xx.xxx -p xxxx`

Connection attempts were then verified through the Cowrie logs.

5. Limitations & Risks

- Misconfigured VM networking could expose the honeypot outside the environment.
- Lack of monitoring beyond logs might make it difficult to catch issues during long-running tests.
- Simple setup that will not work on advanced attackers, reducing realism and accuracy of the results.

6. Testing & Findings

6.1 Automated Password Guessing Using Hydra

A password-guessing attack was simulated using a small password list:

- 123
- password
- admin
- root
- test

The attack was executed using Hydra: `hydra -l root -P passwords.txt ssh://192.xxx.xx.xxx -s xxxx`

Findings

```
New connection: 192.xxx.xx.xxx:32850
New connection: 192.xxx.xx.xxx:32864
New connection: 192.xxx.xx.xxx:32866
New connection: 192.xxx.xx.xxx:32888
New connection: 192.xxx.xx.xxx:32896
login attempt [b'root'/b'password'] succeeded
login attempt [b'root'/b'admin'] succeeded
login attempt [b'root'/b'test'] succeeded
login attempt [b'root'/b'123'] succeeded
login attempt [b'root'/b'root'] failed
```

- Multiple rapid connection attempts were observed from the attacker IP (192.xxx.xx.xxx), indicating automated brute-force activity.
- Various password combinations were tested against the root account.
- Several login attempts were successfully accepted by the honeypot due to the use of weak credentials.

```

$ hydra -l root -P passwords.txt ssh://192.168.1.100 -s 2222
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military
or secret service organizations, or for illegal purposes (this is non-binding, these
*** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-19 16:59:14
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommend
ed to reduce the tasks: use -t 4
[DATA] max 5 tasks per 1 server, overall 5 tasks, 5 login tries (l:1/p:5), ~1 try per
task
[DATA] attacking ssh://192.168.1.100:2222/
[2222][ssh] host: 192.168.1.100 login: root password: password
[2222][ssh] host: 192.168.1.100 login: root password: admin
[2222][ssh] host: 192.168.1.100 login: root password: test
[2222][ssh] host: 192.168.1.100 login: root password: 123
1 of 1 target successfully completed, 4 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-19 16:59:15

```

Figure 2 – Attacker: password brute-force

```

2026-04-19T16:59:13.623230Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 192.168.1.100]
2026-04-19T16:59:13.628261Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 192.168.1.100]
2026-04-19T16:59:13.629015Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 192.168.1.100]
2026-04-19T16:59:13.629866Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 192.168.1.100]
2026-04-19T16:59:13.631669Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 192.168.1.100]

```

Figure 3 – Logs: connections

```

2026-04-19T16:59:13.667732Z [HoneyPotSSHTransport,2,192.168.1.100] login attempt [b'root'/b'password'] succeeded
2026-04-19T16:59:13.668424Z [HoneyPotSSHTransport,3,192.168.1.100] login attempt [b'root'/b'admin'] succeeded
2026-04-19T16:59:13.668889Z [HoneyPotSSHTransport,4,192.168.1.100] login attempt [b'root'/b'test'] succeeded
2026-04-19T16:59:13.672415Z [HoneyPotSSHTransport,1,192.168.1.100] login attempt [b'root'/b'123'] succeeded
2026-04-19T16:59:13.672995Z [HoneyPotSSHTransport,5,192.168.1.100] login attempt [b'root'/b'root'] failed

```

Figure 4 – Logs: login attempts

6.2 Manual Common-Credential Testing

Manual login attempts were performed using commonly used usernames:

- root: `ssh root@xxx.xxx.xx.xxx -p xxxx`
- admin: `ssh admin@xxx.xxx.xx.xxx -p xxxx`
- user: `ssh user@xxx.xxx.xx.xxx -p xxxx`

Each account was tested with the following passwords:

Username	Password Tested
root	root, test, password
admin	admin, 123, password
user	user, 123, password

Findings

```
New connection: 192.xxx.xx.xxx:xxxxx
[session: xx7aec83a5ae]

login attempt [b'root'/b'root'] failed
login attempt [b'root'/b'test'] succeeded

New connection: 192.xxx.xx.xxx:xxxxx
[session: c8dee3d11472]

login attempt [b'admin'/b'admin'] failed
login attempt [b'admin'/b'123'] failed
login attempt [b'admin'/b'password'] failed

New connection: 192.xxx.xx.xxx:xxxxx
[session: 22328958558e]

login attempt [b'user'/b'user'] failed
login attempt [b'user'/b'123'] failed
login attempt [b'user'/b'password'] failed
```

```

(jin@jin)-[~]
$ ssh root@192.168.1.100 -p 22
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@192.168.1.100:~#
Permission denied, please try again.
root@192.168.1.100:~#
Permission denied, please try again.

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@192.168.1.100:~# exit
Connection to 192.168.1.100 closed.

(jin@jin)-[~]
$ ssh admin@192.168.1.100 -p 22
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
admin@192.168.1.100:~#
Permission denied, please try again.
admin@192.168.1.100:~#
Permission denied, please try again.
admin@192.168.1.100:~#
Permission denied, please try again.
admin@192.168.1.100:~#
Permission denied (publickey,password).

(jin@jin)-[~]
$ ssh user@192.168.1.100 -p 22
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
user@192.168.1.100:~#
Permission denied, please try again.
user@192.168.1.100:~#
Permission denied, please try again.
user@192.168.1.100:~#
Permission denied, please try again.
user@192.168.1.100:~#
Permission denied (publickey,password).

```

Figure 5 – Attacker: login attempts

```

2026-04-19T17:21:30.329922Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 1]
[REDACTED]

2026-04-19T17:21:38.685788Z [HoneyPotSSHTransport,0,192.168.1.100] Login attempt [b'root'/b'root'] failed
[REDACTED]

2026-04-19T17:21:41.968150Z [HoneyPotSSHTransport,0,192.168.1.100] Login attempt [b'root'/b'test'] succeeded
[REDACTED]

2026-04-19T17:22:39.147466Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.100 (192.168.1.100) [session: 2]
[REDACTED]

2026-04-19T17:22:41.438801Z [HoneyPotSSHTransport,1,192.168.1.100] Login attempt [b'admin'/b'admin'] failed
[REDACTED]

2026-04-19T17:22:44.432073Z [HoneyPotSSHTransport,1,192.168.1.100] Login attempt [b'admin'/b'123'] failed
[REDACTED]

2026-04-19T17:22:47.097163Z [HoneyPotSSHTransport,1,192.168.1.100] Login attempt [b'admin'/b'password'] failed
[REDACTED]

```

```
2026-04-19T17:22:56.105991Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168. (192. ) [session: ]  
[REDACTED]  
2026-04-19T17:22:57.722953Z [HoneyPotSSHTransport,2,192. ] login attempt [b'user'/b'user'] failed  
[REDACTED]  
2026-04-19T17:22:59.793264Z [HoneyPotSSHTransport,2,192. ] login attempt [b'user'/b'123'] failed  
[REDACTED]  
2026-04-19T17:23:02.356579Z [HoneyPotSSHTransport,2,192. ] login attempt [b'user'/b'password'] failed  
[REDACTED]
```

Figure 6, 7 & 8 – Logs: login attempts by attacker

6.3 Successful Login and Command Execution

A successful login was performed using weak credentials:

```
ssh root@192.xxx.xx.xxx -p xxxx
```

```
Password: test
```

After gaining access, the following commands were executed:

Command	Observation
<code>uname -a</code>	Retrieves basic system information.
<code>whoami</code>	Identifies the current user and privilege level.
<code>ls</code>	Lists files in the current directory.
<code>cat /etc/passwd</code>	Displays user account information.

This test demonstrates attacker behaviour after gaining access to a system.

Findings

```
New connection: 192.xxx.xx.xxx:xxxxx
[session: 0bc72c71e1cb]

2026-04-19T17:38:28.860277Z
[HoneyPotSSHTransport,0,192.xxx.xx.xxx] CMD: uname -a

2026-04-19T17:38:31.849205Z
[HoneyPotSSHTransport,0,192.xxx.xx.xxx] CMD: whoami

2026-04-19T17:38:33.590691Z
[HoneyPotSSHTransport,0,192.xxx.xx.xxx] CMD: ls

2026-04-19T17:38:37.810867Z
[HoneyPotSSHTransport,0,192.xxx.xx.xxx] CMD: cat /etc/passwd
```

A successful connection was established by the attacker IP (192.xxx.xx.xxx). The commands demonstrate typical attacker reconnaissance behaviour.

```

(jin@jin)-[~]
$ ssh root@192.168.1.1 -p 2222
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@192.168.1.1's password:
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@192.168.1.1:~# uname -a
Linux 192.168.1.1 3.2.0-4-amd64 #1 SMP Debian 3.2.68-1+deb7u1 x86_64 GNU/Linux
root@192.168.1.1:~# whoami
root
root@192.168.1.1:~# ls
root@192.168.1.1:~# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
sshd:x:101:65534::/var/run/sshd:/usr/sbin/nologin
root@192.168.1.1:~#

```

Figure 9 – Attacker: command execution

```

2026-04-19T17:38:17.337211Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.1.1 (192.168.1.1) [session: 192.168.1.1]
2026-04-19T17:38:18.832994Z [HoneyPotSSHTransport,0,192.168.1.1] login attempt [b'root'/b'test'] succeeded

```

Figure 10 – Logs: login successful

```

2026-04-19T17:38:18.852182Z [twisted.conch.ssh.session#info] Getting shell
2026-04-19T17:38:28.860277Z [HoneyPotSSHTransport,0,192.168.1.1] CMD: uname -a
2026-04-19T17:38:28.861414Z [HoneyPotSSHTransport,0,192.168.1.1] Command found: uname -a
2026-04-19T17:38:31.849205Z [HoneyPotSSHTransport,0,192.168.1.1] CMD: whoami
2026-04-19T17:38:31.849671Z [HoneyPotSSHTransport,0,192.168.1.1] Command found: whoami
2026-04-19T17:38:33.590691Z [HoneyPotSSHTransport,0,192.168.1.1] CMD: ls
2026-04-19T17:38:33.592613Z [HoneyPotSSHTransport,0,192.168.1.1] Command found: ls
2026-04-19T17:38:37.810867Z [HoneyPotSSHTransport,0,192.168.1.1] CMD: cat /etc/passwd
2026-04-19T17:38:37.815978Z [HoneyPotSSHTransport,0,192.168.1.1] Command found: cat /etc/passwd

```

Figure 11 – Logs: command execution

6.4 Test Summary

Test	Result	Main Findings
Automated Password Guessing Using Hydra	Passed	Rapid automated login attempts were captured by Cowrie.
Manual Common-Credential Testing	Passed	root/test was accepted while the tested admin and user credentials failed.
Successful Login and Command Execution	Passed	Cowrie captured post-login commands and attacker interaction.

7. Challenges

Challenge	Resolution
Initial lack of planning and understanding resulted in constant switching between NAT and Host-Only Adapter modes to install required dependencies.	Improved understanding of when each network configuration was required during setup.
Shared clipboard functionality between the host and Ubuntu VM did not initially work, affecting workflow efficiency.	Resolved by installing VirtualBox Guest Additions and enabling bidirectional clipboard support.
Dependency installation issues were encountered due to environment inconsistencies and package version conflicts.	Resolved by using a clean Python virtual environment and updating pip.
Difficulty setting up the Cowrie directory and configuration, including the missing <code>cowrie.cfg</code> file.	Resolved by copying the default configuration template.

8. Improvements

- Improve initial planning to reduce the need for switching between different network configurations, such as NAT and Host-Only modes.
- Implement automated log analysis, such as real-time log visualisation, to filter relevant information and identify patterns in attack behaviour for more efficient analysis.
- Improve the honeypot to include additional services such as HTTP or FTP to capture a wider range of attack behaviours.
- Configure a custom user credentials database to simulate more realistic login scenarios and improve the accuracy of captured interactions.